Module 2.1: Critical Thinking: Claims & Arguments - Key Terms & Definitions

Critical Thinking: Terms & Definitions

Argument: An attempt to support or prove a contention by providing a reason for accepting it. The contention itself is called the conclusion; the statement (or statements) offered as support or demonstration is referred to as the premise (or premises).

- Alternative Definition: **Argument**: A group of statements in which some of them (the premises) are intended to support another of them (the conclusion).
- **Important:** Illustrations, reports, explanations, lists of facts, photos, conditionals, and expository passages **are not arguments!**

Argument from Analogy: An argument that something has an attribute in view of the fact that a similar thing has that attribute.

- **Example:** *Forcing Americans to buy health insurance for their own good is like forcing them to eat broccoli for their own good. It would be wrong to force people to eat broccoli for their own good; therefore, it would be wrong to force them to buy health insurance for their own good.*

Causal Argument: Causal arguments inductively support a claim that asserts or implies cause-and-effect, or, alternatively, use such a cause-and-effect claim as a premise in an argument to establish that something happened or is the case.

- **Example:** *The toilet is leaking. It isn't possible for the toilet to leak without damaging the floor. Therefore, the floor is damaged.*

Claim: When a belief (judgment, opinion) is asserted in a declarative sentence, the result is a claim or statement.

Cogent: A strong inductive argument with true premises.

Cognitive Bias: A psychological factor that unconsciously affects belief formation.

Conclusion: In an argument, the claim thought to be supported or demonstrated by another claim (or claims) referred to as the premise (or premises).

Conclusion indicator: A word or phrase (e.g., "therefore", "so", "thus") that ordinarily introduces the conclusion of an argument.

Conditional Statement: is made up of two basic parts – the antecedent (the if-part) and the consequent (the then-part). Note: Conditionals are not arguments!

Critical Thinking: is thinking that critiques – specifically, evaluating reasoning according to standards. Put differently, we think critically when we rationally evaluate our own or others’ thinking.

Deductive Argument: An argument intended to provide logically conclusive support for its conclusion. Put differently: An argument intended to prove or demonstrate, rather than merely support, a conclusion.

- **Example:** *Dorothy is an engineer. All engineers are good at math. So Dorothy is good at math.*

Evidence: is data deployed to support a reason.

Explanation: A claim or set of claims intended to make another claim, object, event, or state of affairs intelligible. Also, an account of how an object, event, or state of affairs came to be.

“Fact vs. opinion”: Sometimes people refer to true objective claims as “facts,” and use the word “opinion” to designate any claim that is subjective. **But note**: Some opinions are not subjective judgments, because their truth or falsity is independent of what people think.

- ***Example:*** *If you believe that Portland, Oregon, is closer to the North Pole than to the equator, that opinion happens to be true, and would continue to be true even if you change your mind,*

Factual claim (belief/opinion): An objective claim (belief/opinion).

Fallacy: A common mistake in reasoning; a specious argument. Proposing or accepting such an argument both count as fallacies.

Generalizing from a Sample: Reasoning that all, most, or some percentage of the members of a population have an attribute because all, most, or some percentage of a sample of the population have that attribute.

- **Example:** *I've liked the majority of Professor Moore's lectures so far; therefore, probably I will like the majority of all of Professor Moore's lectures.*

Inductive argument: An argument in which the premises are intended to provide probable, not conclusive, support for its conclusion.

- **Example:** *Most engineers are good at math. Larry Page is an engineer. Therefore, Larry is probably good at math.*

Inference: The mental process of reasoning from a premise or premises to a conclusion based on those premises.

Inference to the Best Explanation: An argument that supports a conclusion about what the cause of some outcome is or was, or alternatively an argument that something exists (or did exist) because it is the likeliest cause of some outcome.

- **Example:** *There is water on the floor. The most likely cause is a leaking toilet. Therefore, the toilet is leaking.*

Invalid argument: A deductively argument that isn't valid.

Logic: The branch of philosophy concerned with the principles and criteria of inference.

Moral relativism: The view that what is morally right and wrong depends on and is determined by one's group or culture.

Moral subjectivism: The idea that what is right and wrong is merely a matter of subjective opinion, that thinking something is right or wrong makes it right or wrong for that individual.

Objective Claim: A statement that is not made true or false by the speaker or writer's thinking it is true or false. In other words, *w*hether it is true or false is independent of whether you or anyone else thinks it is true or false.

Opinion: A claim that somebody believes to be true.

Premise: The claim or claims in an argument thought to support or demonstrate a conclusion.

Premise indicator: A word or phrase (e.g., “since it is the case that . . .”, “because”, “since”) that ordinarily introduces a premise of an argument.

Report: A kind of nonargument consisting of one or more statements that convey information about some topic or event.

Sound: A deductively valid argument that has true premises.

Statistical syllogism (De-generalizing): The reverse of generalizing from a sample. Concluding that particular members of a population have an attribute because some high proportion of all the population’s members have that attribute.

- **Example:** *Most days that are hot and humid are days with thunderstorms. It is hot and humid today. Therefore, probably there will be a thunderstorm today.*

Stronger/weaker arguments: The more likely the premise of an inductive argument makes the conclusion, the stronger the argument, and the less likely it makes the conclusion, the weaker the argument.

Subjective statement: A statement that is made true or false by the speaker or writer’s thinking it is true or false.

Truth value: If a claim is true, its truth value is “true”; if it is false, its truth value is “false.”

Unstated Premises: An unstated premise is known as an implied premise.

- Example:

Premise: You can't check out books from the library without an ID.

Conclusion: Bill won't be able to check out any books.

Unstated premises are common in real life because sometimes they seem too obvious to need mentioning. The argument "the car is beyond fixing, so we should get rid of it" actually has an unstated premise to the effect that we should get rid of any car that is beyond fixing; but this may seem so obvious to us that we don't bother stating it. Unstated conclusions also are not uncommon, though they are less common than unstated premises.

DEDUCTION, INDUCTION, AND UNSTATED PREMISES

Somebody announces, "Rain is on its way." Somebody else asks how he knows. He says, "There's a south wind." Is the speaker trying to demonstrate rain is coming? Probably not. His thinking, spelled out, is probably something like this:

Stated premise: The wind is from the south.

Unstated premise: Around here, south winds are usually followed by rain.

Conclusion: There will be rain.

In other words, the speaker was merely trying to show that rain was a good possibility. Notice, though, that the unstated premise in the argument could have been a **universal statement** to the effect that a south wind always is followed by rain at this particular location, in which case the argument would be deductive:

Stated premise: The wind is from the south.

Unstated premise: Around here, a south wind is always followed by rain.

Conclusion: There will be rain.

Spelled out this way, the speaker's thinking is deductive: It isn't possible for the premises to be true and the conclusion to be false. So one might wonder abstractly what the speaker intended—an inductive argument that supports the belief that rain is coming, or a deductive demonstration.

There is, perhaps, no way to be certain short of asking the speaker something like, "Are you 100 percent positive?" But experience ("background knowledge") tells us that wind from a particular direction is not a surefire indicator of rain. So probably the speaker did have in mind merely the first argument. He wasn't trying to present a 100 percent certain, knock-down demonstration that it would rain; he was merely trying to establish there was a good chance of rain.

You can always turn an inductive argument with an unstated premise into a deductively valid argument by supplying the right universal premise—a statement that something holds without exception or is true everywhere or in all cases. Is that what the speaker really has in mind, though? You have to use background knowledge and common sense to answer the question.

For example, you overhear someone saying,

Stacy and Justin are on the brink of divorce. They're always fighting. One could turn this into a valid deductive argument by adding to it the universal statement "Every couple fighting is on the brink of divorce."

But such an unqualified universal statement seems unlikely. Probably the speaker wasn't trying to demonstrate that Stacy and Justin are on the brink of divorce. He or she was merely trying to raise its likelihood. He or she was presenting evidence that Stacy and Justin are on the brink of divorce.

Often it is clear that the speaker does have a deductive argument in mind and has left some appropriate premise unstated. You overhear Professor Greene saying to Professor Brown, **"Flunk her! This is the second time you've caught her cheating."**

It would be strange to think that Professor Greene is merely trying to make it more likely that Professor Brown should flunk the student. Indeed, it is hard even to make sense of that suggestion. Professor Greene's argument, spelled out, must be this: "Flunk her! This is the second time you've caught her cheating." So context and content often make it clear what unstated premise a speaker has in mind and whether the argument is deductive or inductive.

Unfortunately, though, this isn't always the case. We might hear someone say,

Stated premise: This is the second time you've caught her cheating.

Unstated premise: Anyone who has been caught cheating two times should be

flunked.

Conclusion: She should be flunked.

The bars are closed; therefore, it is later than 2 A.M. If the unstated premise in the speaker's mind is something like "In this city, the bars all close at 2 A.M.," then presumably he or she is thinking deductively and is evidently proffering proof that it's after 2. But if the speaker's unstated premise is something like "Most bars in this city close at 2 A.M." or "Bars in this city usually close at 2 A.M.," then we have an inductive argument that merely supports the conclusion. So which is the unstated premise? We really can't say without knowing more about the situation or the speaker.

Valid argument: An argument such that it would be self-contradictory to maintain that the conclusion is false and the premise (or premises) are true. Put differently, if the premises are true, the conclusion <u>must</u> follow.

Warrant: is the "bridge"/"connection"; the general principle connecting claims and reasons.